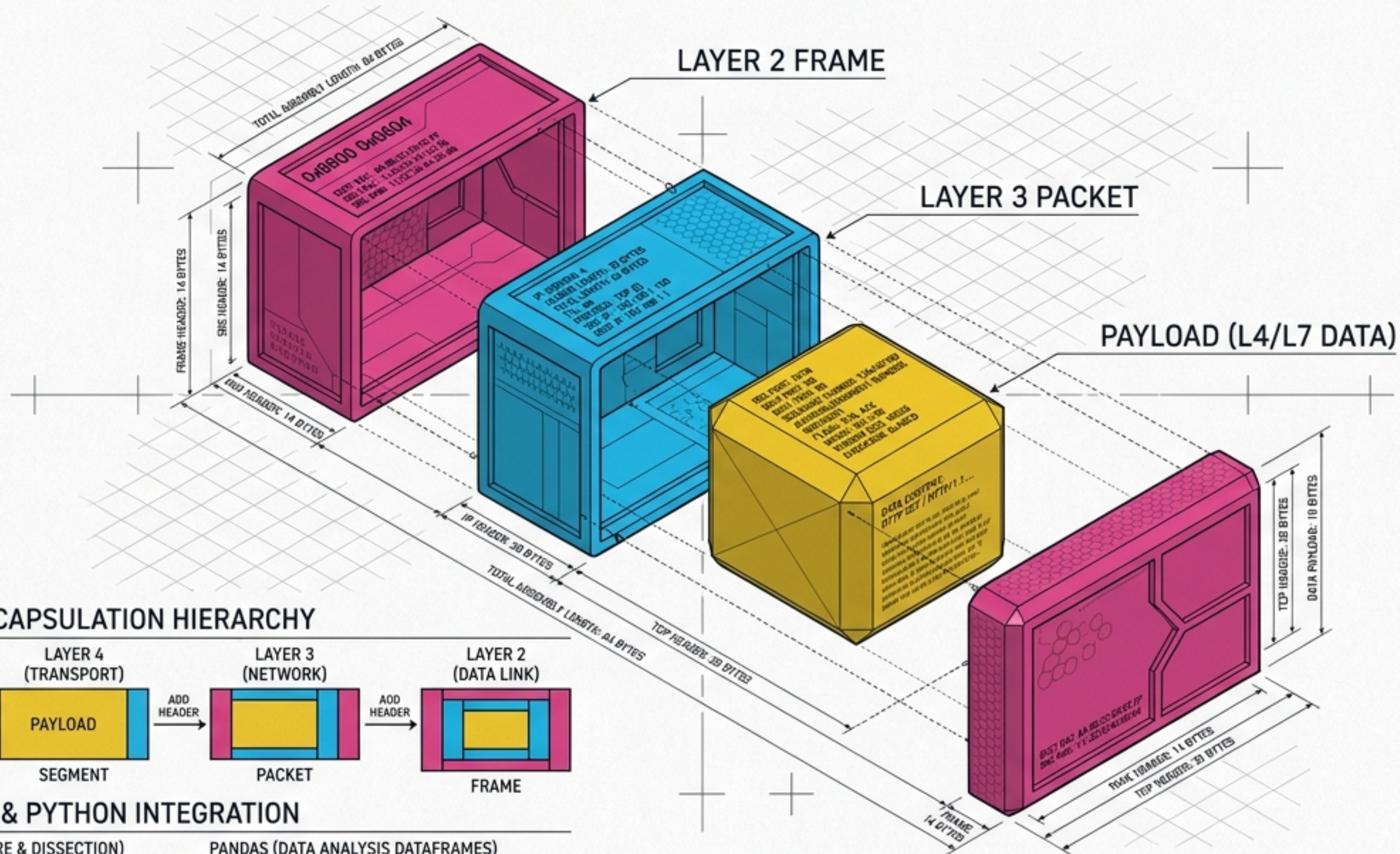


LAB 1.4: PACKETS, FRAMES, AND DATA FLOW

PROVING NETWORK THEORY THROUGH EMPIRICAL DATA, PACKET ANALYSIS, AND PYTHON.



ANALYSIS TOOLS & PYTHON INTEGRATION

WIRESHARK (PACKET CAPTURE & DISSECTION)
TCPDUMP (CLI PACKET ANALYZER)
SCAPY (PYTHON PACKET MANIPULATION LIBRARY)

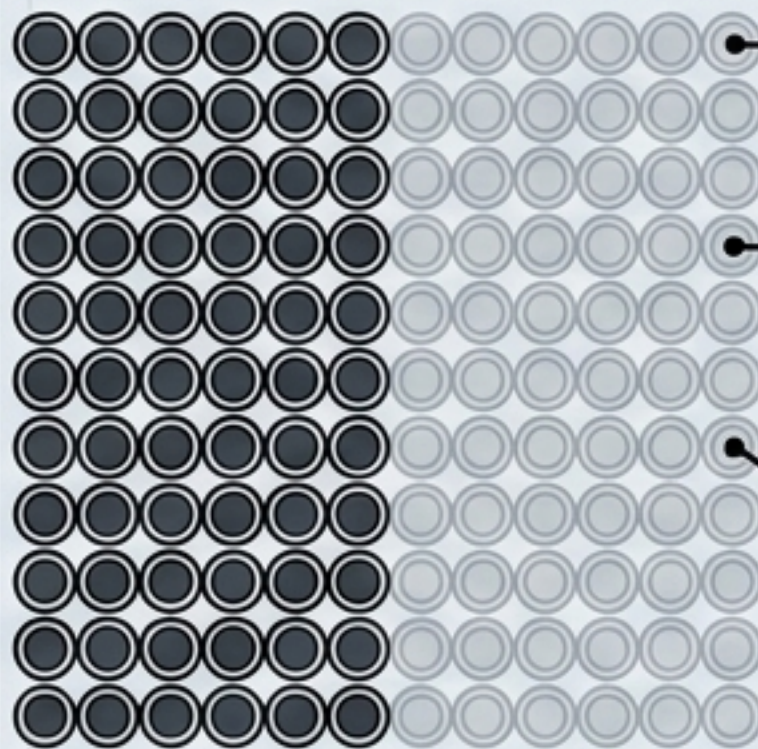
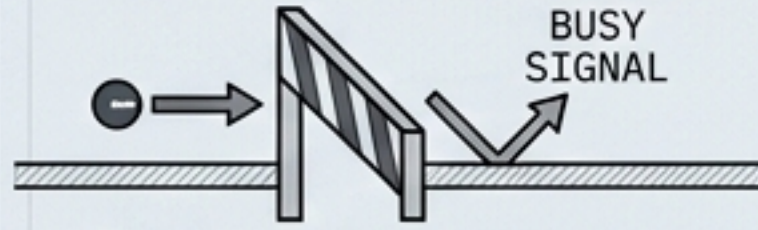
PANDAS (DATA ANALYSIS DATAFRAMES)
MATPLOTLIB/SEABORN (DATA VISUALIZATION)

The Evidence Agenda

1	Statistical Multiplexing	Measuring switching efficiency.
2	Packet Independence	Tracing dynamic routing.
3	Hop-by-Hop Rebuilding	Verifying frame reconstruction via TTL.
4	Path MTU	Triggering and observing fragmentation.
5	The Overhead Tax	Calculating Bandwidth vs. Goodput.

The Scale Multiplier: Circuit vs. Packet Switching

CIRCUIT SWITCHING

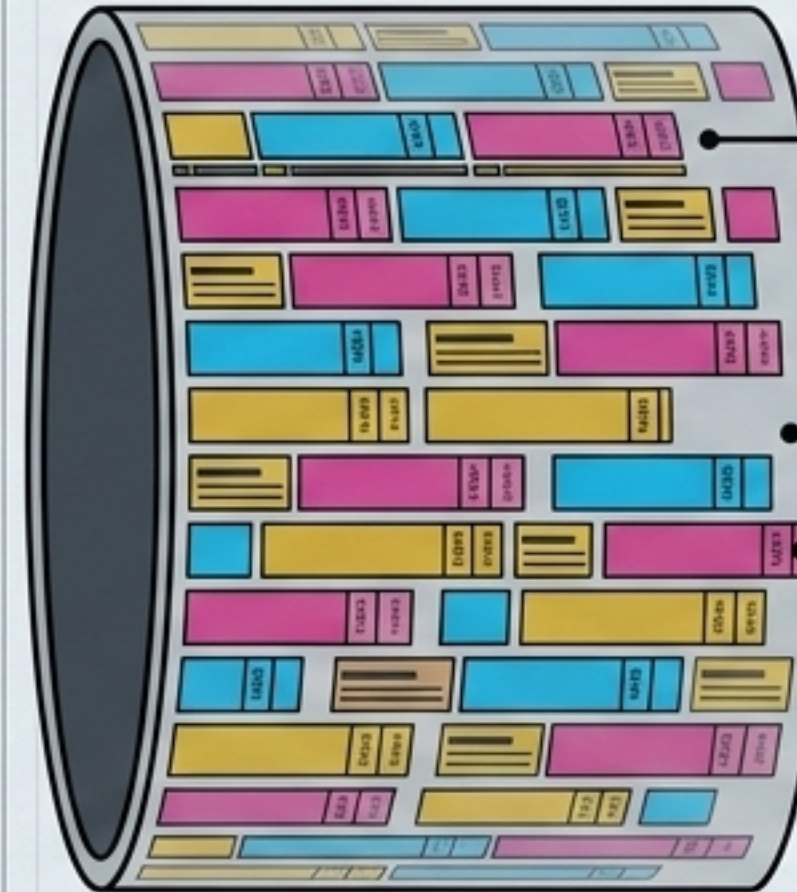


100 users max.

60% bandwidth wasted when idle.

A hard limit on capacity resulting in busy signals.

PACKET SWITCHING



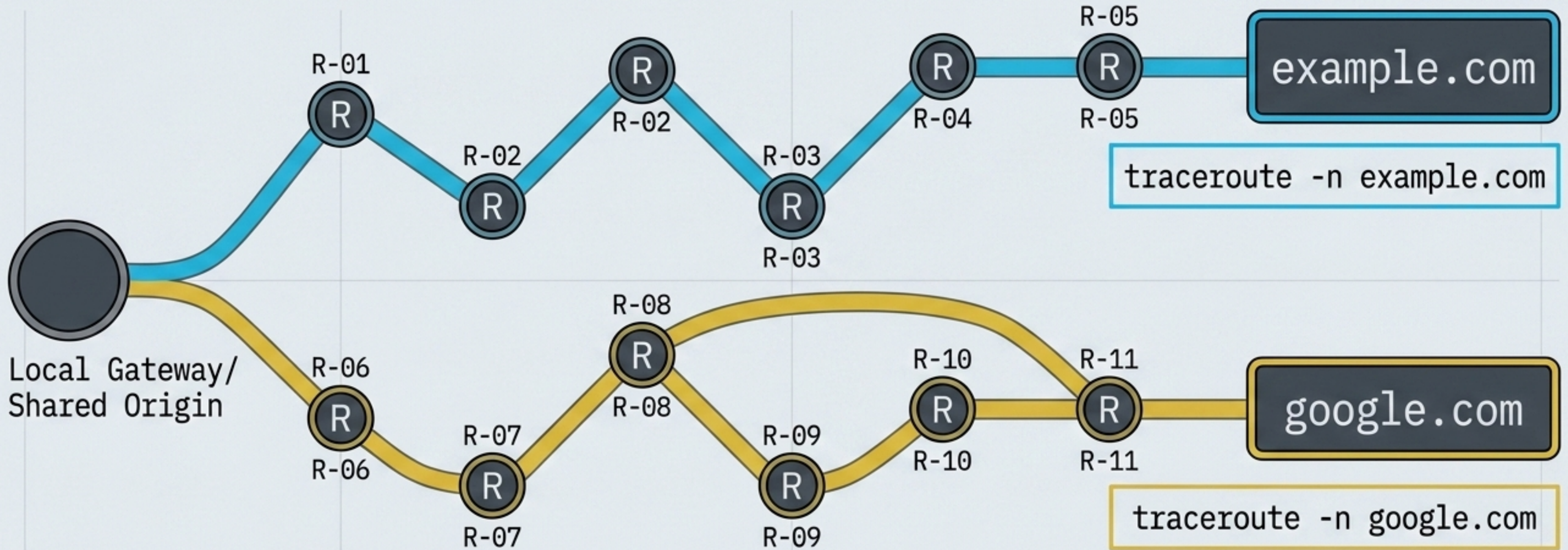
1000 users supported (active 10% of the time).

Idle uses zero bandwidth.

Congestion risk is mathematically negligible (1 in millions).

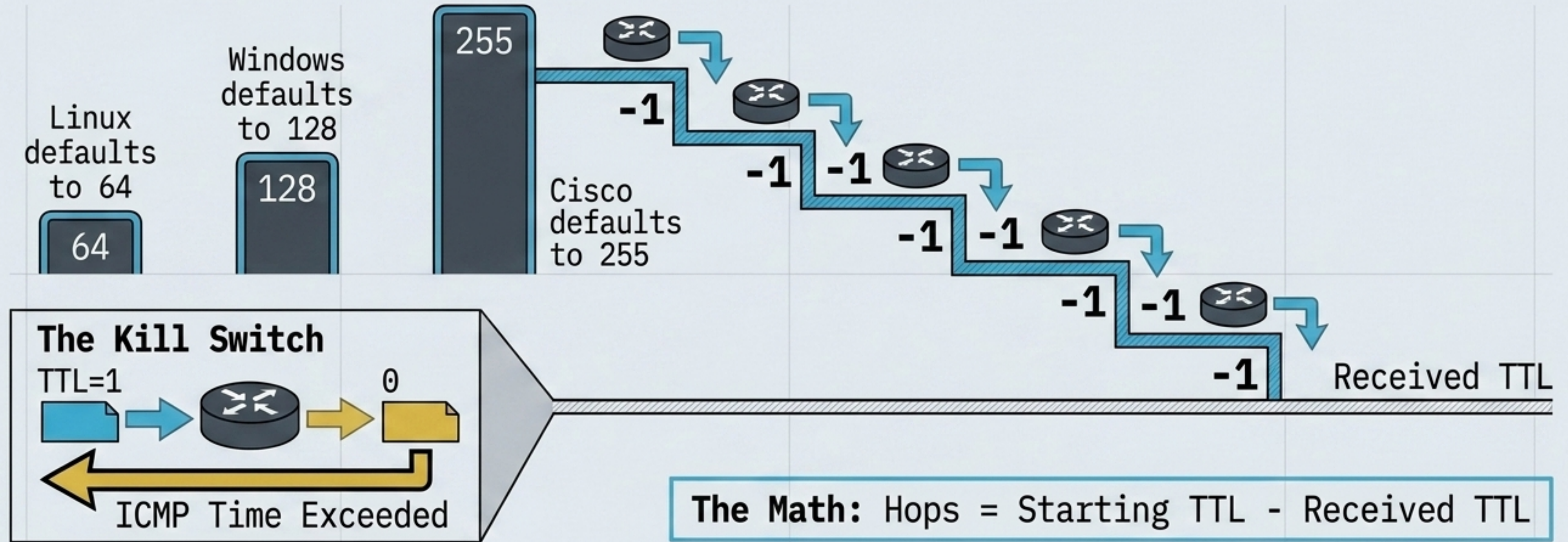
Insight: Packet switching trades guaranteed, dedicated capacity for massive scale via statistical multiplexing.

Packet Independence: No Reserved Paths



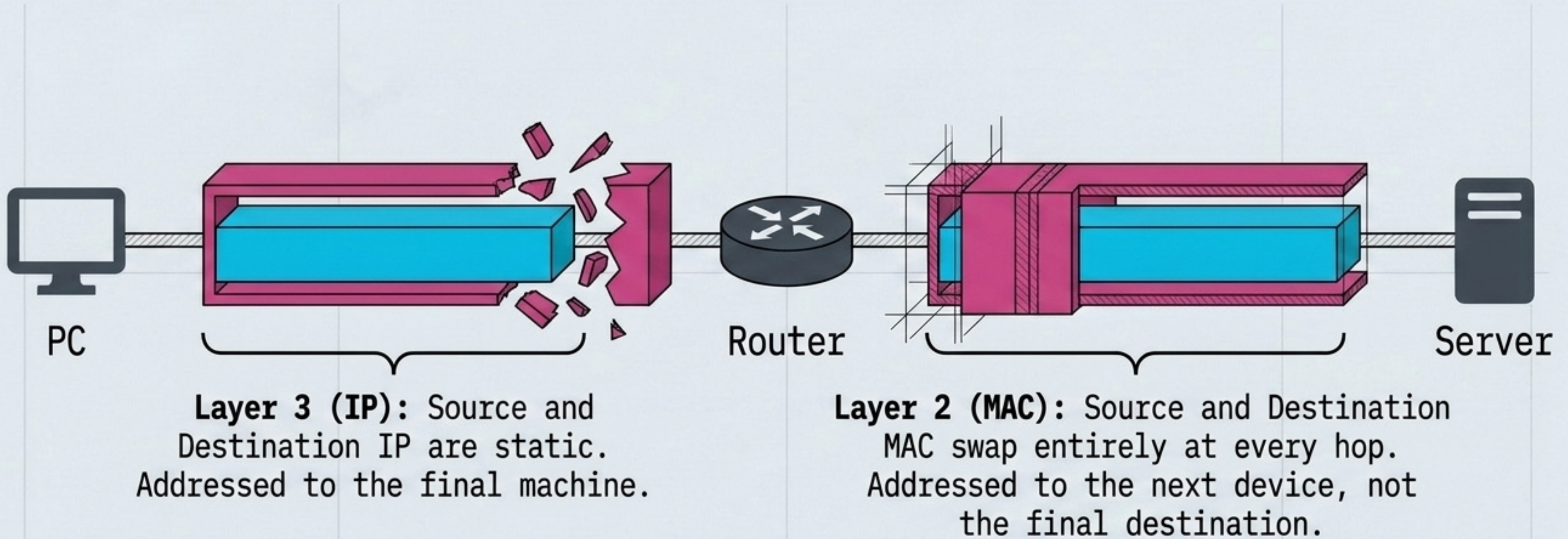
Insight: Packets share the same local gateway but diverge wildly across the network. No single route is reserved; the network routes dynamically hop-by-hop.

Counting Hops via TTL Decrement



Insight: Every single router physically touches and modifies the IP packet by reducing its lifespan.

Hop-by-Hop: The Illusion of a Single Frame

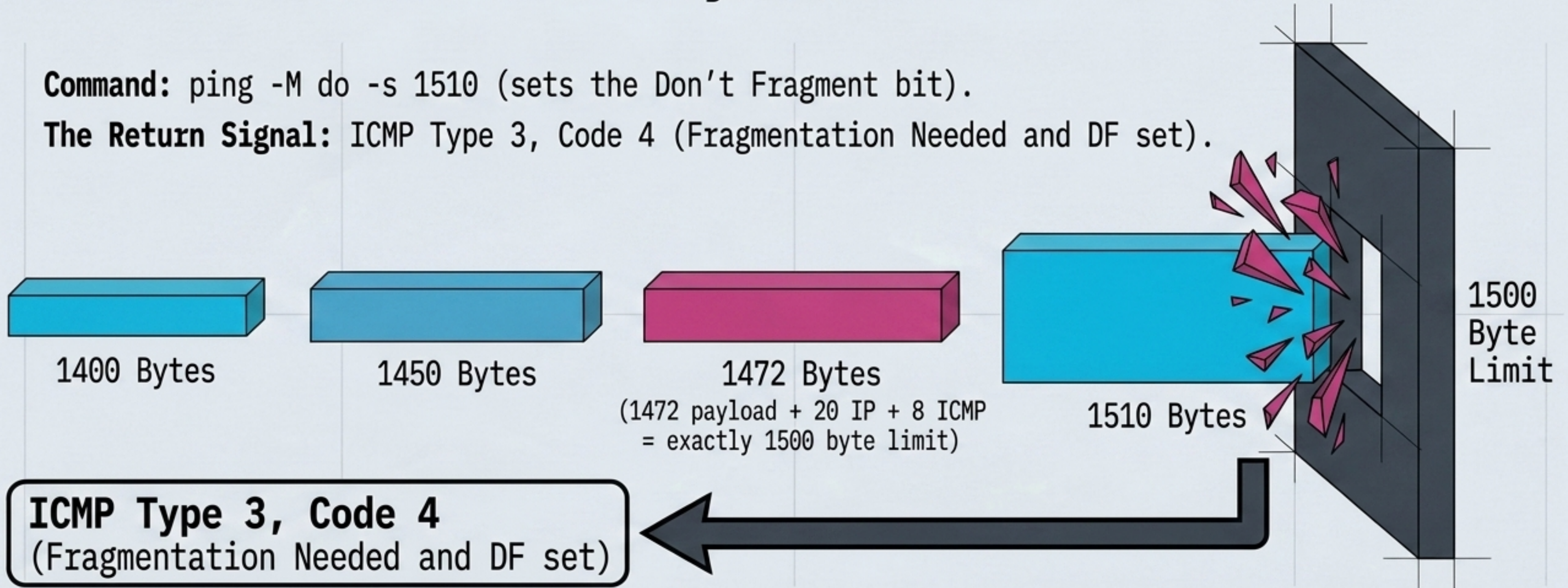


Insight: Routers strip the incoming frame and build an entirely new one. MAC addresses are strictly local; IP addresses are global end-to-end.

Path MTU Discovery (PMTUD)

Command: `ping -M do -s 1510` (sets the Don't Fragment bit).

The Return Signal: ICMP Type 3, Code 4 (Fragmentation Needed and DF set).

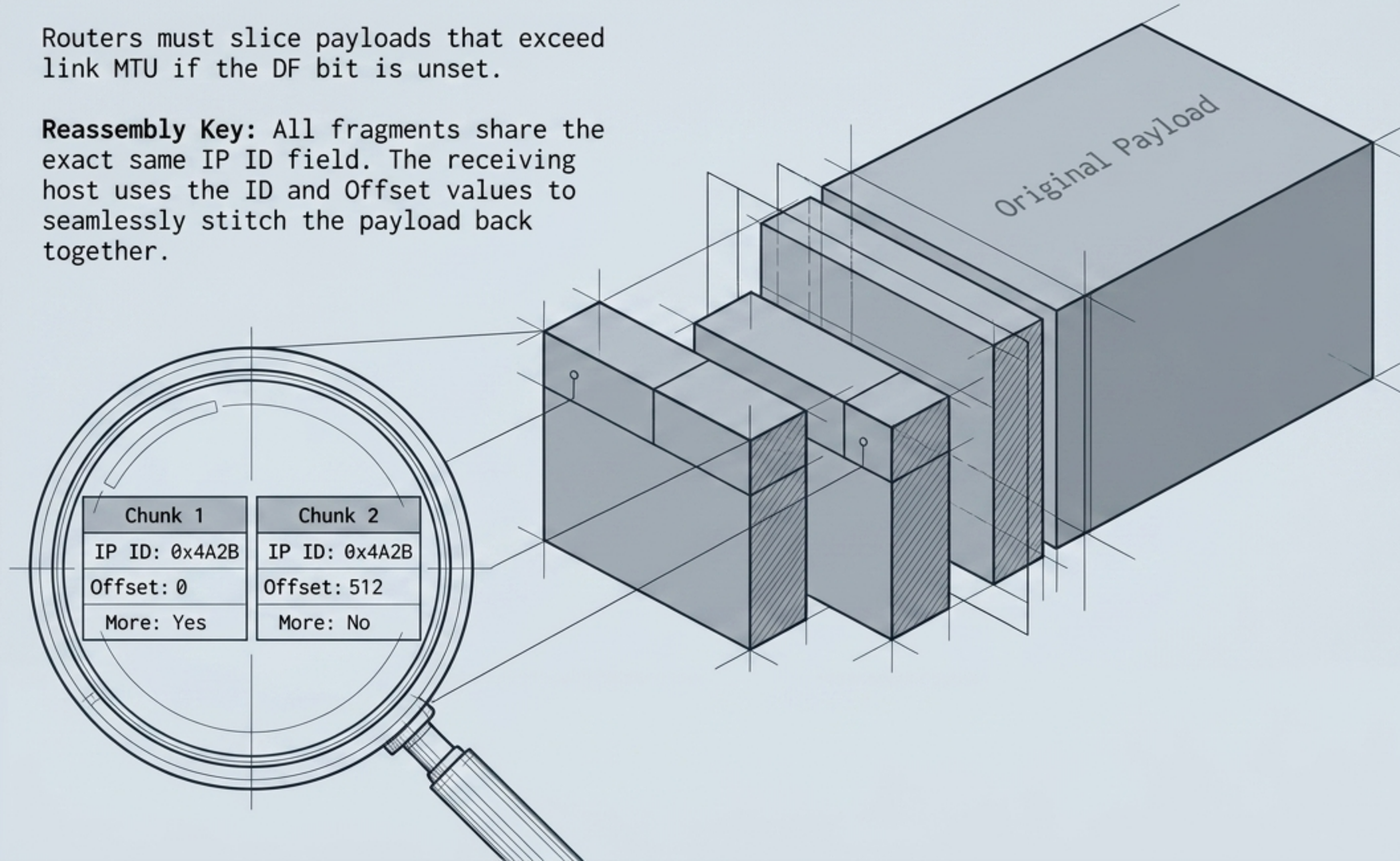


Insight: When a packet is too large and marked Do Not Fragment, the blocking router sends an ICMP error telling the sender the exact maximum size it can handle.

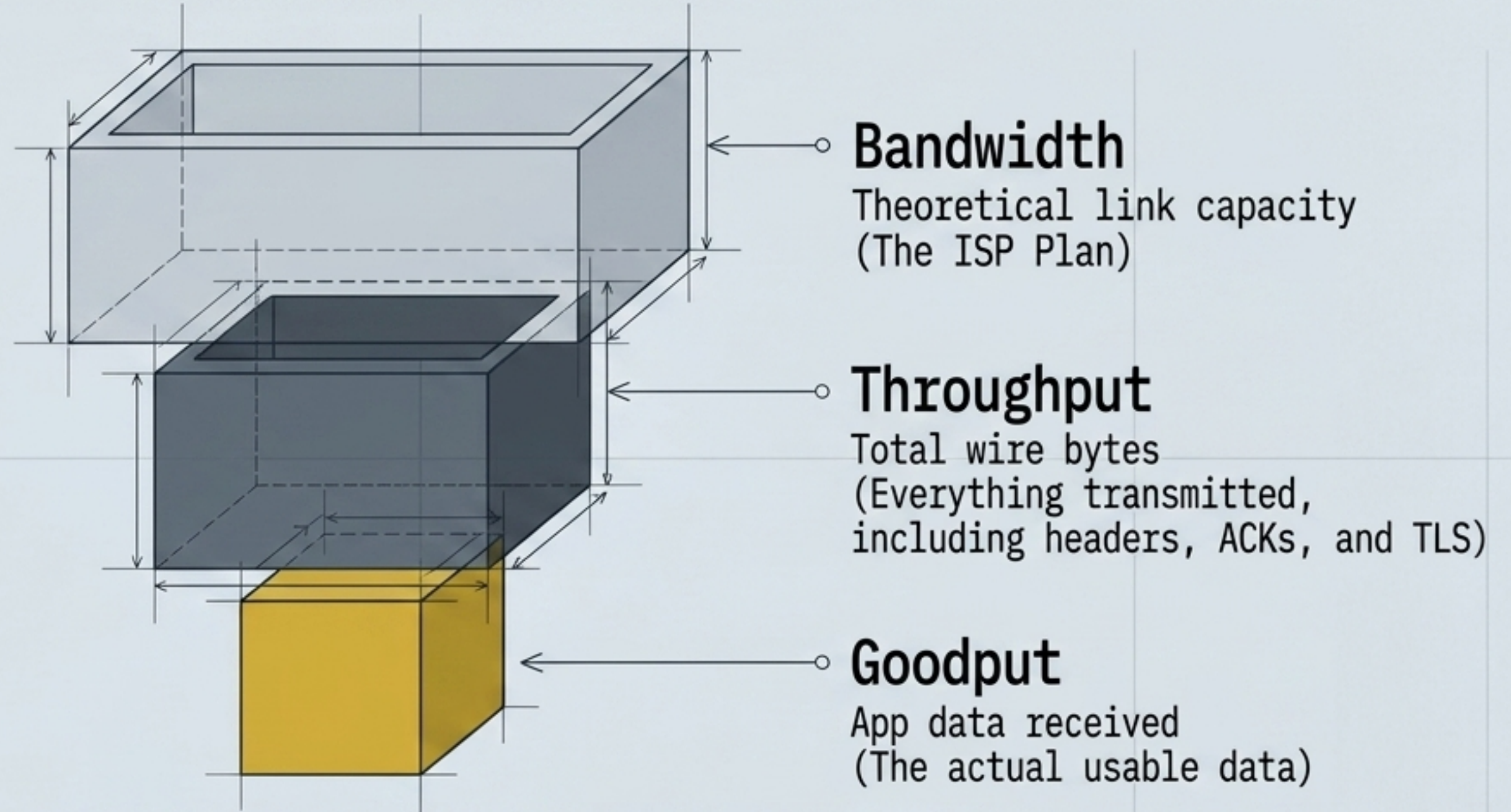
The Anatomy of IPv4 Fragmentation

Routers must slice payloads that exceed link MTU if the DF bit is unset.

Reassembly Key: All fragments share the exact same IP ID field. The receiving host uses the ID and Offset values to seamlessly stitch the payload back together.

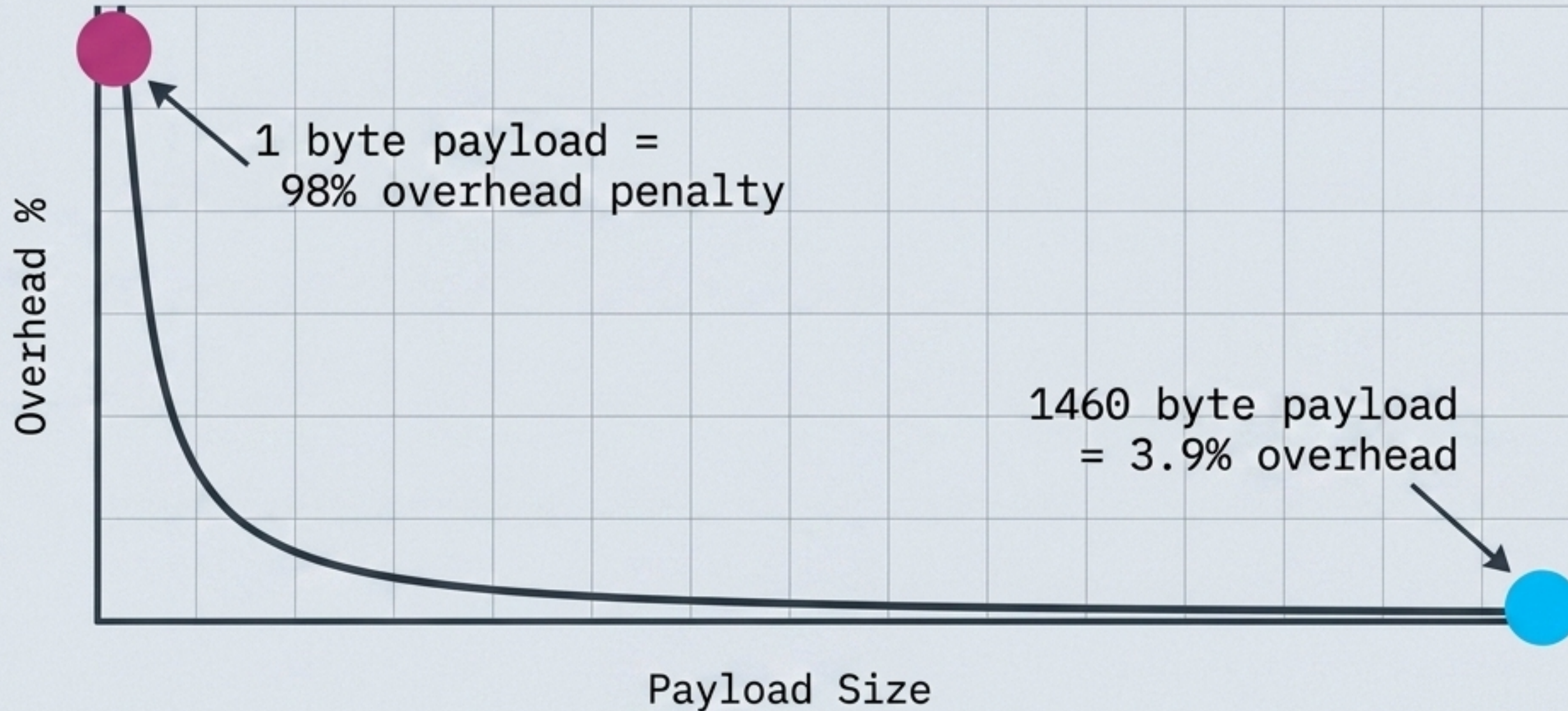


The Efficiency Funnel: Bandwidth vs. Goodput



Insight: $\text{Goodput} \leq \text{Throughput} \leq \text{Bandwidth}$. The unavoidable gap is the tax of protocol overhead. A 500 Mbps connection never delivers 500 Mbps of actual data.

The Protocol Overhead Tax



The fixed tax: 54+ bytes of headers (Ethernet 14 + FCS 4 + IP 20 + TCP 20).

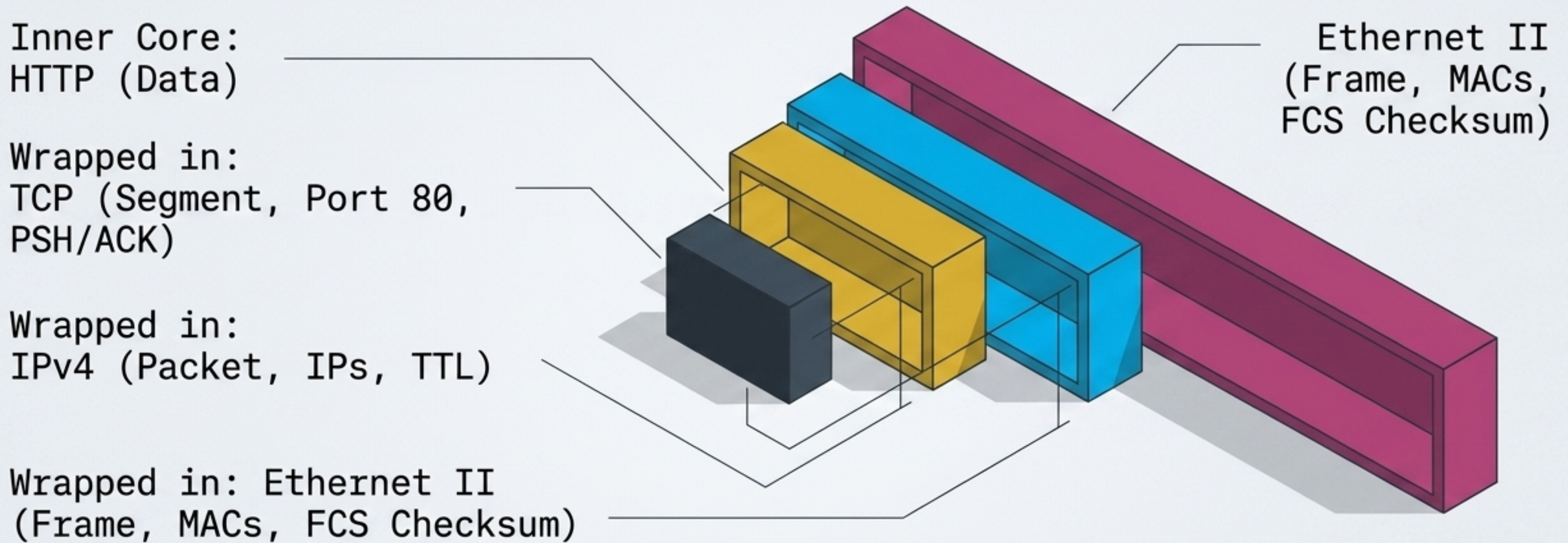
Insight: Applications that send tiny, fragmented messages choke network efficiency. Maximizing payload minimizes the fixed header tax.

Mapping the Address Tuple

	Layer	Identifies	Scope
Source/Dest MAC	Layer 2	The NIC	Local Network
Source/Dest IP	Layer 3	The Machine	End-to-End
Source/Dest Port	Layer 4	The Process/Service	End-to-End

The Spotlight: Destination MAC targets the local router, NOT the final remote server.

The Encapsulation Stack



Insight: Each layer wraps the layer above it, adding its own specific control information. The Frame is the largest data unit because it literally carries everything else across the physical wire.

Code to Concept: Parsing Raw Packets

FF	FF	FF	FF	FF	FF	AB	CD	EF	01	23	45	08
00	45	00	00	00	54	00	00	40	00	40	06	00
00	C0	A8	00	01	C0	A8	00	01	C0	A8	00	0A
C0	A8	00	01	06	89	0E	0E	00	13	23	45	08
00	00	00	00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	01	03	00	00	00	00
00	00	00	00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	01	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	00	00	00	00
00	00	00	00	00	00	00	00	00	3C	20	1D	8F
FF	FF	FF	FB	EB	EF	FF	AB	CD	00	00	00	00
00	00	00	00	00	2F	00	F1	01	23	20	00	00
00	00	00	00	00	00	00	00	00	00	00	00	00
00	00	00										

```
data = bytes.fromhex("FFFFFFFFFFFFFFF  
FFFABCDEF012345080045000054...")
```

```
# Extract Destination MAC
```

```
dest_mac = data[0:6]
```

```
# Extract Source IP
```

```
source_ip = data[12:16]
```

Wireshark isn't magic.

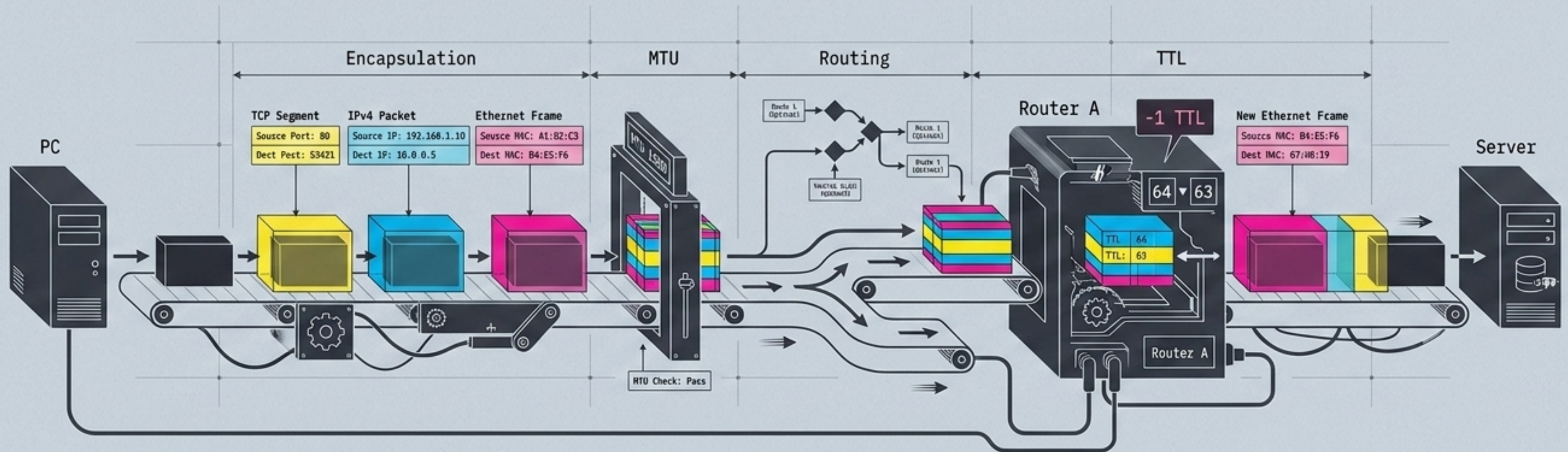
`data[0:6]` extracts the Destination MAC.

`data[12:16]` extracts the Source IP.

Insight: To a computer, a network frame is just an array of bytes. Protocols are just rigid agreements on where specific data starts and stops.

Anatomy of a Data Flow

Insight: The illusion of a seamless internet connection is maintained by thousands of microscopic, hop-by-hop mechanical translations happening in milliseconds. A packet is a dynamic, morphing envelope constantly negotiating its environment.



Evidence Verified: Lab Takeaways

- ✓ Measured statistical multiplexing vs. circuit limits.
- ✓ Traced hop-by-hop MAC address rebuilding via captures.
- ✓ Triggered PMTUD and identified ICMP limits.
- ✓ Calculated the Goodput protocol tax for varied payloads.
- ✓ Parsed strict layer offsets directly from raw binary.

**Theoretical networking models verified
through empirical packet analysis**